

Faculty Event Management System

Group Project — Workload Division (4 Members)

Spring Boot · Thymeleaf · MySQL · Maven

M1

User & Security

M2

Society & Membership

M3

Events & Booking

M4

Venues, Approval & Reports

Member 1 — User & Security Management

- User registration, login, and logout
- User CRUD (create, read, update, deactivate accounts)
- Role management — ADMIN / SOCIETY_ADMIN / MEMBER
- BCrypt password hashing (BCryptPasswordEncoder)
- SecurityConfig.java — URL-level access rules
- Role-specific dashboard view (/dashboard)

Key Files: AuthController · UserService · SecurityConfig · UserRepository
Templates: auth/login, auth/register, dashboard/dashboard

Member 2 — Society Management & Membership

- Society CRUD (create, read, update, deactivate)
- Assign and change society admin user
- Add and remove members from a society
- Society member list view
- society_members table management

Key Files: SocietyController · SocietyService · SocietyRepository SocietyMemberRepository · **Templates:** societies/*

Member 3 — Event Management & Venue Booking

- Event CRUD — create, read, update, delete (only before booking approved)
- Venue booking request form (/bookings/new)
- Conflict detection logic — interval-overlap JPQL query ★ Core Feature
- BookingConflictException handling and error display on form
- My Bookings page — society admin view (/bookings/my)

Key Files: EventController · BookingController · VenueBookingService VenueBookingRepository · Templates: events/*, bookings/form, bookings/list

Member 4 — Venue Management, Approval & Reports

- Venue CRUD (create, read, update, deactivate) — venues table
- Booking approval / rejection workflow — admin view (/admin/bookings/pending)
- Re-check conflict at approval time (race condition guard)
- Email notifications on approve/reject (EmailService + JavaMailSender)
- Venue availability calendar view (/calendar)
- Booking history page (/admin/bookings/history)
- Export report — PDF/CSV (/admin/reports/export)

Key Files: VenueController · AdminBookingController · CalendarController EmailService · AdminReportController ·
Templates: venues/*, admin/*, calendar/*

Corrections from Original Split

"Approve Events"	Events are not approved — venue bookings are approved by the admin.	Renamed to Booking Approval Workflow, assigned to Member 4.
Venue Management alone	Only one table (venues) — too light compared to other members.	Paired with booking approval, email notifications, calendar, and reports.
Conflict detection unassigned	The most critical feature of the system had no clear owner.	Explicitly assigned to Member 3 alongside the booking request flow.
Member 3 overloaded	Had 7 responsibilities — nearly double the others.	Booking approval + email + reports moved to Member 4 to balance load.

Note: All members must register an account, then the first admin manually sets **role = 'ADMIN'** in the database. Member 3 owns the most technically complex feature (conflict detection). If workload feels uneven during development, the CSV/PDF export from Member 4 can be swapped with Member 3's event delete feature.